ANSWER KEY & MARKING SCHEME — MATHEMATICS · CLASS 10 · SET 01

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION – A (Multiple Choice Questions · 20 × 1 = 20) — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions. Q19–Q20: a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

1. If the HCF of 65 and 117 is expressible in the form $65m - 117$, then the value of m is

(a) 2 (b) 1 (c) 3 (d) 4

Ans: (a) HCF(65, 117) = 13. 65m − 117 = 13 ⇒ 65m = 130 ⇒ m = 2.

2. If one zero of the quadratic polynomial $x^2 + 3x + k$ is 2, then the value of k is

(a) 10 (b) 5 (c) −5 (d) −10

Ans: (d) p(2) = 4 + 6 + k = 0 ⇒ k = −10.

3. The pair of linear equations $x + 2y - 5 = 0$ and $-3x - 6y + 15 = 0$ has

(a) a unique solution (b) exactly two solutions
(c) infinitely many solutions (d) no solution

Ans: (c) $a_1/a_2 = b_1/b_2 = c_1/c_2 = -1/3$ (coincident lines).

4. The values of k for which the equation $2x^2 + kx + 3 = 0$ has equal roots are

(a) ± √6 (b) ± 2√6
(c) ± 24 (d) 6, −6 only if k > 0

Ans: (b) D = $k^2 - 24 = 0$ ⇒ k = ±√24 = ±2√6.

5. Which of the following quadratic equations has two distinct real roots?

(a) $x^2 + x + 1 = 0$ (b) $2x^2 - 6x + 3 = 0$
(c) $x^2 - 4x + 4 = 0$ (d) $3x^2 + 2x + 1 = 0$

Ans: (b) D = 36 − 24 = 12 > 0 for the correct option; D = −3, 0, −8 for the others.

6. The 10th term from the end of the AP 4, 9, 14, …, 254 is

(a) 199 (b) 204 (c) 209 (d) 214

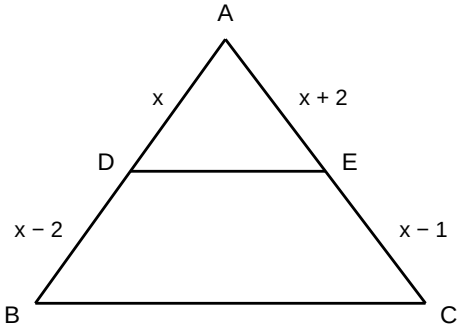
Ans: (c) 10th from end = l − 9d = 254 − 45 = 209.

7. The nth term of an AP is $3n + 5$. The sum of its first 10 terms is

(a) 215 (b) 185 (c) 230 (d) 245

Ans: (a) a = 8, d = 3; $S_{10} = 5(16 + 27) = 215$.

8. In the given figure, DE ∥ BC. If AD = x, DB = $x - 2$, AE = $x + 2$ and EC = $x - 1$, then the value of x is



(a) 3 (b) 5 (c) 6 (d) 4

Ans: (d) $x/(x-2) = (x+2)/(x-1) \Rightarrow x^2 - x = x^2 - 4 \Rightarrow x = 4$ (BPT).

9. $\Delta ABC \sim \Delta PQR$ and $ar(\Delta ABC) : ar(\Delta PQR) = 9 : 16$. If AB = 6 cm, then PQ equals

(a) 8 cm (b) 4.5 cm (c) 9 cm (d) 10.5 cm

Ans: (a) $AB/PQ = \sqrt{(9/16)} = 3/4 \Rightarrow PQ = 8$ cm.

10. The point on the *x*-axis which is equidistant from (3, 2) and (−5, −2) is

(a) (1, 0) (b) (0, −1) (c) (−1, 0) (d) (−2, 0)

Ans: (c) $(x-3)^2 + 4 = (x+5)^2 + 4 \Rightarrow -6x + 9 = 10x + 25 \Rightarrow x = -1$.

11. If $4\tan\theta = 3$, then $\dfrac{4\sin\theta - \cos\theta}{4\sin\theta + \cos\theta}$ equals

(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{3}{4}$ (d) $\frac{1}{2}$

Ans: (d) Divide by $\cos\theta$: $(4\tan\theta - 1)/(4\tan\theta + 1) = (3 - 1)/(3 + 1) = 1/2$.

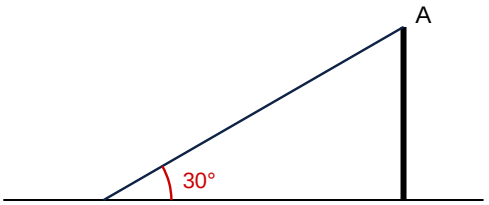
12. Which of the following statements is INCORRECT?

(a) $\sin 30° = \cos 60°$ (b) $\sec 60° = 2\cos 60°$

(c) $\tan 45° \cdot \cot 45° = 1$ (d) $\sin^2 45° + \cos^2 45° = 1$

Ans: (b) $\sec 60° = 2$ whereas $2\cos 60° = 1$.

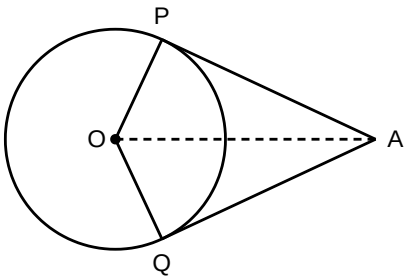
13. The angle of elevation of the top of a tower from a point on the ground, 30 m away from its foot, is 30°. The height of the tower is



(a) $10\sqrt{3}$ m (b) $30\sqrt{3}$ m (c) 15 m (d) 10 m

Ans: (a) $h = 30\tan 30° = 30/\sqrt{3} = 10\sqrt{3}$ m.

14. In the figure, AP and AQ are tangents from A to a circle with centre O. If $\angle PAQ = 80°$, then $\angle POA$ is



(a) 40° (b) 80° (c) 50° (d) 100°

Ans: (c) $\angle POQ = 180° - 80° = 100°$; OA bisects it $\Rightarrow \angle POA = 50°$.

15. The perimeter of a square of side 22 cm is equal to the circumference of a circle. The radius of the circle is

(a) 7 cm (b) 22 cm (c) 28 cm (d) 14 cm

Ans: (d) $2\pi r = 88 \Rightarrow r = 88 \times 7/44 = 14$ cm.

16. A solid sphere of radius 3 cm is melted and recast into small spheres of radius 1 cm. The number of small spheres formed is

(a) 9 (b) 27 (c) 18 (d) 81

Ans: (b) $n = (3)^3/(1)^3 = 27$.

17. For the distribution below, the sum of the lower limits of the median class and the modal class is

Class	0–10	10–20	20–30	30–40	40–50
Frequency	6	14	12	18	10

(a) 40 (b) 60 (c) 50 (d) 70

Ans: (c) N = 60, N/2 = 30; cf: 6, 20, 32 ⇒ median class 20–30. Highest f = 18 ⇒ modal class 30–40. 20 + 30 = 50.

18. Two coins are tossed together. The probability of getting at most one head is

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) 1 (d) $\frac{3}{4}$

Ans: (d) Outcomes HH, HT, TH, TT; at most one head: HT, TH, TT = 3/4.

19. **Assertion (A):** The probability that a leap year selected at random has 53 Sundays is 2/7.
Reason (R): A leap year has 52 weeks and 2 extra days; the pair of extra days can occur in 7 equally likely ways, 2 of which contain a Sunday.

Ans: (a) Both true; R explains A.

20. **Assertion (A):** The quadratic equation $2x^2 - 4x + 3 = 0$ has two distinct real roots.
Reason (R): A quadratic equation has real roots if and only if its discriminant is non-negative.

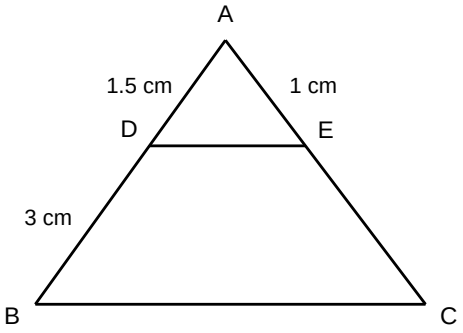
Ans: (d) $D = 16 - 24 = -8 < 0 \Rightarrow$ A is false; R is true.

SECTION – B (Very Short Answer Questions · 5 × 2 = 10) — 5 questions · 10 marks

21. Find the HCF and LCM of 26 and 91 by the prime factorisation method. Hence verify that HCF × LCM = product of the two numbers.

$26 = 2 \times 13$, $91 = 7 \times 13 \Rightarrow$ HCF = 13, LCM = $2 \times 7 \times 13 = 182$ (1).
HCF × LCM = $13 \times 182 = 2366 = 26 \times 91$ ✓ (1).

22. In the figure (DE ∥ BC), AD = 1.5 cm, DB = 3 cm and AE = 1 cm. Find EC.



By BPT: AD/DB = AE/EC (1) ⇒ 1.5/3 = 1/EC ⇒ EC = 2 cm (1).

OR

The diagonals of a trapezium ABCD (AB ∥ DC) intersect at O. If AB = 2 CD, find the ratio of the areas of ΔAOB and ΔCOD.

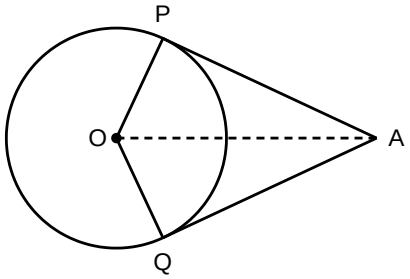
> ΔAOB ~ ΔCOD (AA) (1). Ratio of areas = $(AB/CD)^2 = 4 : 1$ (1).

23. Two concentric circles have radii 7 cm and 25 cm. Find the length of the chord of the larger circle which touches the smaller circle.

> Half-chord² = $25^2 - 7^2 = 576$ (1) ⇒ half-chord = 24 ⇒ chord = 48 cm (1).

OR

In the figure, AP and AQ are tangents from A to a circle with centre O and ∠PAQ = 70°. Find ∠OPQ.



> ∠POQ = 180° − 70° = 110° (1). OP = OQ ⇒ ∠OPQ = (180° − 110°)/2 = 35° (1).

24. Find the zeros of the polynomial $6x^2 - 7x - 3$ and verify the relationship between the zeros and the coefficients.

> $6x^2 - 9x + 2x - 3 = (2x - 3)(3x + 1)$ ⇒ zeros 3/2, −1/3 (1). Sum = 7/6 = −(−7)/6 ✓; product = −1/2 = −3/6 ✓ (1).

25. A bag contains 5 red, 3 white and 2 black balls. A ball is drawn at random. Find the probability that it is (i) white (ii) neither red nor white.

> Total = 10. (i) P(white) = 3/10 (1). (ii) neither red nor white = black: 2/10 = 1/5 (1).

SECTION – C (Short Answer Questions · 6 × 3 = 18) — 6 questions · 18 marks

26. Find the ratio in which the *y*-axis divides the line segment joining A(−4, −6) and B(10, 12). Also find the point of division.

> Let ratio be k : 1; x-coordinate = (10k − 4)/(k + 1) = 0 (1) ⇒ k = 2/5 i.e. 2 : 5 (1).
> y = (2 × 12 + 5 × (−6))/7 = −6/7. Point = (0, −6/7) (1).

27. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact.

> Let tangent XY touch circle (centre O) at P; take Q ≠ P on XY. Q lies outside the circle ⇒ OQ > OP (1). This holds for every Q ≠ P, so OP is the shortest distance from O to XY (1) ⇒ OP ⊥ XY (1).

OR

Prove that the lengths of the two tangents drawn from an external point to a circle are equal.

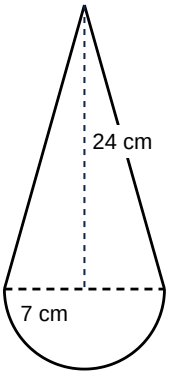
> Tangents PA, PB from P; join OA, OB, OP. OA ⊥ PA, OB ⊥ PB (1). In ΔOAP and ΔOBP: OA = OB (radii), OP common, ∠OAP = ∠OBP = 90° (1) ⇒ RHS congruence ⇒ PA = PB (1).

28. The daily wages (in ₹ hundreds) of 35 workers are given. Find the mean wage.

Wage	10–20	20–30	30–40	40–50	50–60
No. of workers	4	7	12	9	3

> Mid-values 15, 25, 35, 45, 55; Σf = 35 (½). Σfx = 60 + 175 + 420 + 405 + 165 = 1225 (1½). Mean = 1225/35 = 35 (hundreds) = ₹ 3500 (1).

29. A toy is in the form of a cone of radius 7 cm and height 24 cm mounted on a hemisphere of the same radius. Find the total surface area of the toy. (Use π = 22/7)



Slant height l = √(7² + 24²) = 25 cm (1). TSA = πrl + 2πr² (1) = (22/7)(7)(25) + 2(22/7)(49) = 550 + 308 = 858 cm² (1).

30. Prove that (sin θ + cosec θ)² + (cos θ + sec θ)² = 7 + tan² θ + cot² θ.

LHS = sin²θ + cosec²θ + 2 + cos²θ + sec²θ + 2 (1) = (sin²θ + cos²θ) + 4 + (1 + cot²θ) + (1 + tan²θ) (1) = 1 + 4 + 2 + tan²θ + cot²θ = RHS (1).

O R

If tan θ + sin θ = *m* and tan θ − sin θ = *n*, show that (*m*² − *n*²)² = 16*mn*.

m² − n² = 4 tanθ sinθ (1) ⇒ (m² − n²)² = 16 tan²θ sin²θ. mn = tan²θ − sin²θ = sin²θ(sec²θ − 1) = sin²θ tan²θ (1) ⇒ (m² − n²)² = 16mn (1).

31. A train travels 360 km at a uniform speed. Had the speed been 5 km/h more, it would have taken 1 hour less for the same journey. Find the original speed of the train.

360/x − 360/(x + 5) = 1 ⇒ x² + 5x − 1800 = 0 (1) ⇒ (x + 45)(x − 40) = 0 (1) ⇒ x = 40 km/h (1).

SECTION – D (Long Answer Questions · 4 × 5 = 20) — 4 questions · 20 marks

32. Show that the points A(1, 7), B(4, 2), C(−1, −1) and D(−4, 4) are the vertices of a square.

AB = √(9 + 25) = √34, BC = √(25 + 9) = √34, CD = √(9 + 25) = √34, DA = √(25 + 9) = √34 (2). All sides equal (1). AC = √(4 + 64) = √68, BD = √(64 + 4) = √68 — diagonals equal (2). Hence ABCD is a square.

O R

Find the coordinates of the points which divide the line segment joining A(−2, 2) and B(2, 8) into four equal parts.

Midpoint of AB: (0, 5) (1). Midpoint of A and (0, 5): (−1, 3.5) (2). Midpoint of (0, 5) and B: (1, 6.5) (2). Points: (−1, 3.5), (0, 5), (1, 6.5).

33. The following table shows the marks of 50 students. Find the median marks.

Marks	0–10	10–20	20–30	30–40	40–50	50–60
No. of students	5	8	15	12	6	4

cf: 5, 13, 28, 40, 46, 50 (2). N/2 = 25 ⇒ median class 20–30 (1). l = 20, cf = 13, f = 15, h = 10 (1). Median = 20 + (25 − 13)/15 × 10 = 28 (1).

34. Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides.

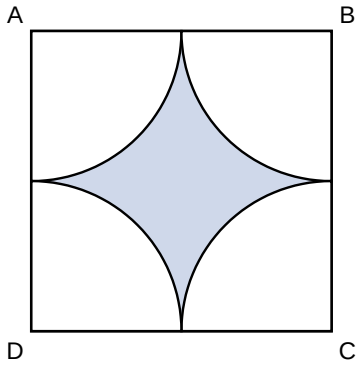
Given ΔABC ~ ΔPQR. Draw AM ⊥ BC, PN ⊥ QR (1). ar(ABC)/ar(PQR) = (½·BC·AM)/(½·QR·PN) = BC·AM/(QR·PN) (1). ΔABM ~ ΔPQN (AA) ⇒ AM/PN = AB/PQ (1). Also AB/PQ = BC/QR = CA/RP (1). ⇒ ar(ABC)/ar(PQR) = (BC/QR)(AB/PQ) = (AB/PQ)² = (BC/QR)² = (CA/RP)² (1).

OR

In an equilateral triangle ABC, D is a point on side BC such that BD = ⅓ BC. Prove that 9 AD^2 = 7 AB^2.

> Let side = a. Draw AE ⊥ BC; BE = a/2, $AE^2 = 3a^2/4$ (2). DE = BE − BD = a/2 − a/3 = a/6 (1). $AD^2 = AE^2 + DE^2 = 3a^2/4 + a^2/36 = 28a^2/36 = 7a^2/9$ (1) ⇒ $9AD^2 = 7AB^2$ (1).

35. In the figure, ABCD is a square of side 14 cm. With the vertices A, B, C, D as centres, four quadrants of radius 7 cm are drawn (unshaded). Find the area and the perimeter of the shaded region. (Use π = 22/7)



> Area of square = 196 cm^2 (1). Area of 4 quadrants = 4 × ¼ × (22/7) × 49 = 154 cm^2 (2). Shaded area = 196 − 154 = 42 cm^2 (1).
> Perimeter = 4 arcs, each ¼ × 2π × 7 = 11 cm ⇒ 44 cm (1).

SECTION – E (Case Study Based Questions · 3 × 4 = 12) — 3 questions · 12 marks

36. The canteen of Navodaya Public School sells samosas and cups of tea. On Monday, Ritu bought 3 samosas and 2 cups of tea for ₹ 70. On Tuesday, Karan bought 2 samosas and 5 cups of tea for ₹ 76. Let the price of one samosa be ₹ *x* and of one cup of tea be ₹ *y*.

(i) Write the pair of linear equations formed by the given information.

> 3x + 2y = 70 and 2x + 5y = 76 (1).

(ii) Without solving, state with a reason whether the pair has a unique solution.

> $a_1/a_2 = 3/2$, $b_1/b_2 = 2/5$; $a_1/a_2 \neq b_1/b_2$ ⇒ unique solution (1).

(iii) Find the price of one samosa and one cup of tea.

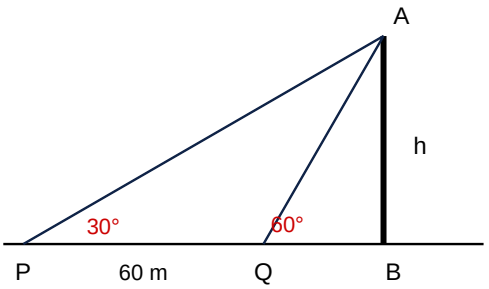
> Multiply (1) by 5, (2) by 2: 15x + 10y = 350, 4x + 10y = 152 ⇒ 11x = 198 ⇒ x = 18 (1½). y = (70 − 54)/2 = 8. Samosa ₹ 18, tea ₹ 8 (½).

OR

(iii) Find the cost of 5 samosas and 3 cups of tea at these prices.

> Solve for x = 18, y = 8 (1). Cost = 5 × 18 + 3 × 8 = 90 + 24 = ₹ 114 (1).

37. The students of Class X measured the height of the school water tank AB using a clinometer. From a point P on the level ground, the angle of elevation of the top A of the tank is 30°. They walk 60 m towards the tank to a point Q, from where the angle of elevation of A is 60°. Let AB = *h* metres. (Take √3 = 1.73)



(i) Express BQ in terms of *h*.

In ΔABQ, tan 60° = h/BQ ⇒ BQ = h/√3 (1).

(ii) Express BP in terms of *h*.

In ΔABP, tan 30° = h/BP ⇒ BP = h√3 (1).

(iii) Find the height of the tank.

BP − BQ = PQ ⇒ h√3 − h/√3 = 60 ⇒ 2h/√3 = 60 (1) ⇒ h = 30√3 = 51.9 m (1).

OR

(iii) Find the distances BQ and BP.

h = 30√3 m. BQ = h/√3 = 30 m (1); BP = h√3 = 90 m (1).

38. Shruti starts a saving plan to buy a bicycle. She saves ₹ 150 in the first month and increases her saving by ₹ 25 every month. Her monthly savings form an AP.

(i) Write the first three terms of the AP and its common difference.

150, 175, 200; d = 25 (1).

(ii) How much does she save in the 12th month?

a_{12} = 150 + 11 × 25 = ₹ 425 (1).

(iii) In which month will she save ₹ 600?

150 + (n − 1)25 = 600 ⇒ n − 1 = 18 (1) ⇒ n = 19, the 19th month (1).

OR

(iii) Find her total savings in the first 12 months.

S_{12} = 12/2 [2(150) + 11 × 25] = 6 × 575 (1) = ₹ 3450 (1).